The ACE worked example — reference-fidelity is not reality-fidelity

Status: computed and guarded — src/pacland/climate/ace_fidelity.py + tests/test_climate_ace_fidelity.py (7 tests, green). This is the citable report the gap analysis flagged as the sharpest single finding (docs/climate-gap-analysis.md §6, target 2).

The finding, in one sentence

> **ACE's "skillful" is a claim about a model's agreement with *another model*, not with the Earth — and its reality-fidelity term is, at best, a ~78% tolerance over 3 observables (or ~96% over the 2 headline observables an atmosphere-only emulator can even reach), not the ">90% skillful" it reports.**

The two-level certificate (what "verification" must mean)

A climate projection is verified only if the bound holds:

```
p_* ≤ p̂_g + p̂_e + radius_s + radius_e
```

term	meaning	ACE's paper computes it?
p̂_g	candidate-vs-oracle — ACE vs its reference GCM (FV3GFS)	computed (~3% of tracked variables)
p̂_e	oracle-vs-truth — the reference GCM vs the Earth	not computed (and for a 100-year run, cannot be)
radius_s	finite-sample term over the scored channels	small (n = 44 channels)
radius_e	sqrt(ln(1/δ_e)/(2m)) over the m reality-validated observables	not computed

ACE's published "verification" is p̂_g alone — a model against a model. The p̂_e term (is the reference GCM itself right against the Earth?) is the whole question for a climate model, and it is absent, exactly as it is for every climate projection.

The ">90%" claim, counted

ACE (arXiv:2310.02074) reports "outperforming a challenging baseline on over 90% of tracked variables." Its Table 1 and Appendix E give the exact numbers:

fact	value	check
output channels	44 = 34 prognostic + 10 diagnostic	channel_summary()
">90% of tracked"	41/44 = 93.2% beat a *2×-coarser FV3GFS baseline	Appendix E: "41 out of 44 output variables"

Every one of the 44 channels is scored against a model — the FV3GFS reference or the coarser FV3GFS baseline. "Reference" and "baseline" are both models, so "skillful" here means *closer to the reference GCM than the coarser GCM is* — reference-fidelity, never reality-fidelity. That is the sentence the two-level certificate makes rigorous.

The reality-fidelity count (the missing `m`)

Map ACE's 44 channels onto physical observables and ask the pacland deficit's question against the Earth (not the reference model): measured (an independent observation c* exists) + computable (ACE outputs O(x)) +

validated (reproduces the historical record within tolerance). The standard is docs/climate-observables.md; the table is ACE_REALITY_OBSERVABLES.

Three honest tiers:

tier	m	radius_e (δ_e = 0.025)	meaning
strict	0	∞ (vacuous)	ACE's own paper validates it against FV3G
anchor	2	0.960	GMST (T7) + EEI (TOA fluxes) — the two c
generous	3	0.784	GMST + EEI + surface pressure (the best-c

ACE is atmosphere-only, so it structurally cannot output the other two of the climate discipline's four anchors — ocean heat content (OHC) and global-mean sea level (GMSL) — they are "measured but not computable by ACE." Even granting ACE full credit for its reference GCM's Earth-validation (the generous tier), the reality certificate radius is ~78%, a tolerance on 3 observables — not the ">90% skillful" headline, which was model-vs-model all along.

Why this is the citable point

- **A "skillful" AI climate model is not a "true" one.** Skill is measured against a reference; truth against the Earth; the AI-emulator frontier currently checks only the first. The two-level certificate is the thing that makes that distinction explicit.
- **It applies one level up from the CMIP6 drift finding.** The conservation guard (docs/climate-cmip6-drift.md) shows the *reference models themselves* leak energy (12 of 19). ACE's verification is fidelity to one of those models. Reference-fidelity to a leaky reference is fidelity to a defect.
- **It names the gap the whole sector inherits.** Oreskes et al. 1994 ("earth models can never be verified, only falsified") stood for 30 years without a number. The deficit m — here 2–3 for the sharpest current emulator — is that number, and it says the field's "verification" certifies a model against another model, not against the Earth.

Honest limits

- **Not a new measurement.** This re-expresses ACE's own published claim (44 channels, 41/44 vs baseline) through the two-level certificate; it does not re-run ACE or FV3GFS. The contribution is the classification and the deficit number, not new model output.
- **`m` is a classification, not a re-analysis.** Each "validated" flag follows docs/climate-observables.md (the four headline observables are validated on the record; the rest are measured-but-not-validated or not measured). A different discipline would re-derive m from its own observable set; the three-tier structure is the durable part.
- **The strict tier is the truth, the generous tier is the concession.** ACE is an *emulator* — "of course it is validated against its reference, not the Earth" is the obvious reply. The generous tier (m = 3, ~78%) is what survives

even after that objection, so it is the headline, not the strict tier.

Reproduce

```
cd pacland
PYTHONPATH=src python3 -m pacland.climate.ace_fidelity       # the summary block
PYTHONPATH=src python3 -m unittest tests.test_climate_ace_fidelity -v   # 7 tests
```

References: ACE, arXiv:2310.02074 ([abstract](https://arxiv.org/abs/2310.02074), [HTML](https://arxiv.org/html/2310.02074v2)); the two-level certificate is docs/verification-formal.md §1–§5.